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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2026

Code: 155612		***Time: 03 Hours***
Semester: VI	***Subject: Python Programming***	***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which of the following statements about indentation in Python is correct?

(i) Indentation is optional and used only for readability
(ii) Indentation determines the scope/block of statements
(iii) Indentation is required only inside functions
(iv) Python uses braces {} instead of indentation

(b)
```
a = 10
b = 3
c = 2
result = a + b * c ** 2
print(result)
```
The value of result is:

(i) 676 (iii) 22
(ii) 28 (iv) 32

(c) Which statement is used to skip the remaining statements of the current iteration of a loop and proceed with the next iteration?

(i) break (iii) pass
(ii) continue (iv) return

(d)
```
for i in range(1, 6):
    if i == 3:
        continue
    print(i, end=" ")
```
What is the output?

(i) 1 2 3 4 5 (iii) 1 2 3
(ii) 1 2 4 5 (iv) 4 5

(e) Which statement about a Python list is correct?

(i) A list cannot contain duplicate elements
(ii) A list is immutable
(iii) A list can contain elements of different data types
(iv) A list can contain only numeric values

(f)
```
L = [10, 20, 30, 40, 50]
print(L[1:4])
```
The output is:

(i) [10, 20, 30] (iii) [20, 30, 40, 50]
(ii) [20, 30, 40] (iv) [10, 20, 30, 40]

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(g) Which statement about a tuple is correct?

(i) Tuples are mutable
(ii) Tuples cannot contain duplicate elements
(iii) Tuples are immutable
(iv) Tuples can contain only numeric values

(h) Which file mode opens a file for reading binary data?

(i) r
(ii) w
(iii) rb
(iv) br

(i) Which OOP concept allows a derived class to acquire properties and methods from a base class?

(i) Polymorphism
(ii) Inheritance
(iii) Compilation
(iv) Iteration

(j) Which of the following is a suitable application of NumPy?

(i) Creating HTML pages
(ii) Numerical and array-based computations
(iii) Sending emails
(iv) Managing operating-system processes only

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Describe the arithmetic and relational operators in Python with suitable examples. Write a Python program to demonstrate their use.

(b) Explain type conversion in Python. Differentiate between implicit and explicit type conversions with suitable code examples

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Write a Python program to find the largest of three numbers using appropriate decision control statements.

(b) Explain the scope and lifetime of variables in Python with suitable examples.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Write a Python program using for and while loops to generate a sequence of numbers from 1 to 10. Explain the working of the loops used.

(b) Write a Python program to demonstrate commonly used string methods such as upper(), lower(), replace(), split() and join().

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Write a Python program to perform basic operations on a list such as adding, removing, updating and accessing elements. Explain the operations used.

(b) Explain list indexing and slicing in Python with suitable examples. Write a Python program to access and extract specific elements from a list using indexing and slicing.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Consider the following dictionary:
student = {"Name": "Shyam", "Age": 18, "Branch": "CSE", "Marks": 85}
Write Python statements to perform different operations such as accessing values, adding a new key-value pair, modifying a value, deleting an item and displaying all keys and values.

(b) Explain the built-in functions and methods available for tuples in Python with suitable examples.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Write a Python program to create a text file, write data into it and read the contents of the file. Explain the file operations used.

(b) Write a Python program to read a text file and count the number of lines, words and characters in the file. Explain the approach used.

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Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Explain Object-Oriented Design in Python. Discuss how OOP concepts are used in designing Python programs.

(b) What is Inheritance? Explain single inheritance in Python with a suitable example.

Q.9 Write short note on any two of the following: **[7 x 2 = 14]**

(a) Operator Precedence and Associativity

(b) zip() Function in Python

(c) Binary Files in Python

(d) NumPy and Its Applications

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